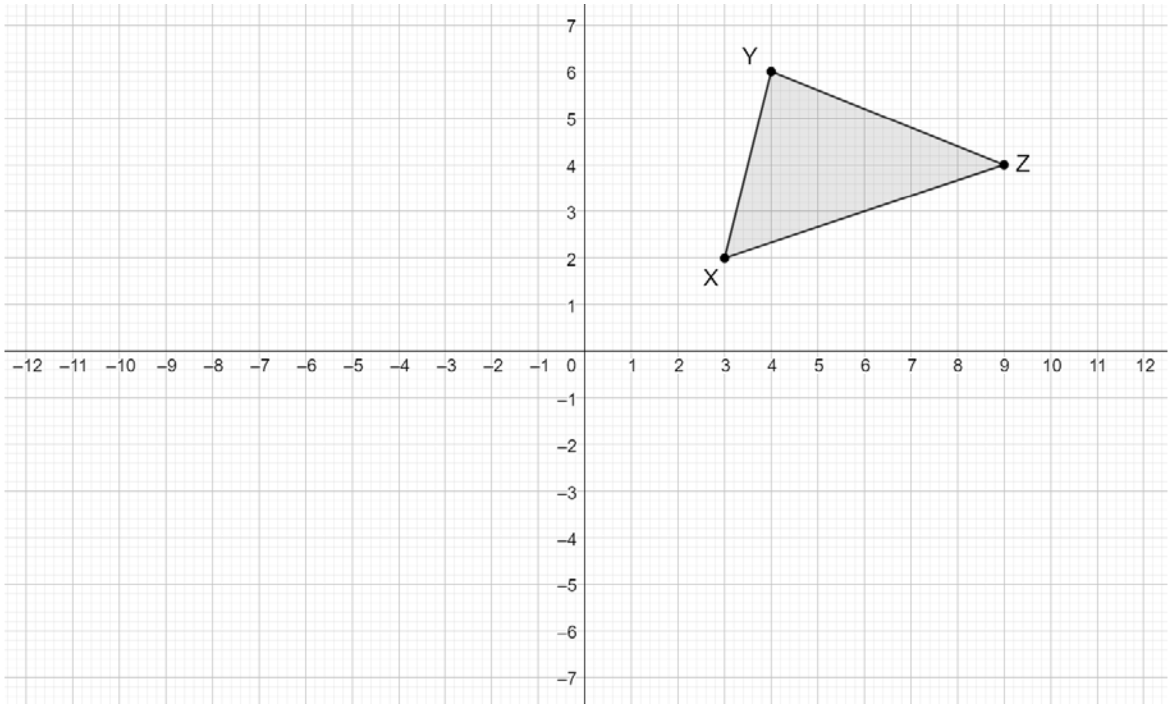
Geometry Name____________________
Unit 6 Date____________________
Lesson H3 Homework Period____________________

$\triangle XYZ$ is shown.



Draw the result of each transformation in the following sequence:

1. $\triangle X'Y'Z'$ as a result of $(x, y) \rightarrow (x - 10, y - 7)$

2. $\triangle X''Y''Z''$ as a result of $(x, y) \rightarrow (x, -y)$

3. $\triangle X'''Y'''Z'''$ as a result of $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$

Indicate the correct relationship for the resulting image:

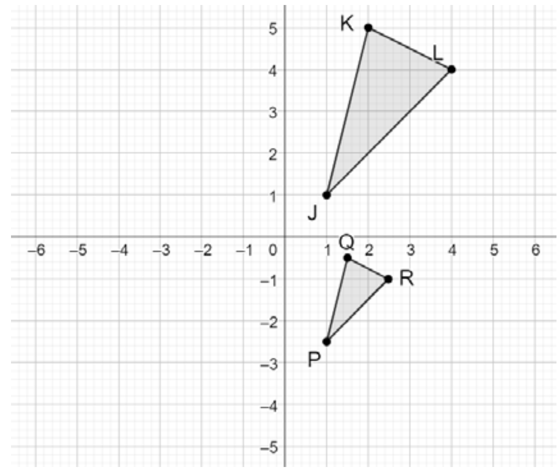
4. Image $\triangle X'Y'Z'$ is [o congruent / o similar] to preimage $\triangle XYZ$.

5. Image $\triangle X''Y''Z''$ is [o congruent / o similar] to preimage $\triangle XYZ$.

6. Image $\triangle X'''Y'''Z'''$ is [o congruent / o similar] to preimage $\triangle XYZ$.

$\triangle JKL$ and $\triangle PQR$ are shown.

Assume: $\frac{JK}{PQ} = \frac{KL}{QR} = \frac{LJ}{RP}$



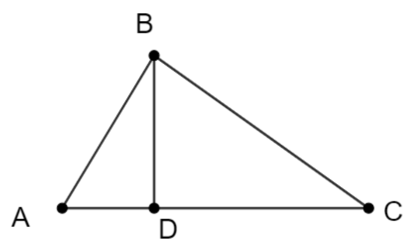
7. Write a series of transformations that will map $\triangle JKL$ onto $\triangle PQR$ so that $\triangle JKL \sim \triangle PQR$.

8. If it is given that $\triangle PQR$ is a result of $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$ then $(x, y) \rightarrow \left(x + \frac{1}{2}, y - 3\right)$,

then the triangles are
- o congruent
- o similar

.

$\triangle ABC$ is shown with $\overline{BD}$.



Complete the table:

	If Given	True or False? $\triangle ABC \sim \triangle ADB$ by AA Similarity	Justification
9.	$\overline{AB} \perp \overline{BC}$ and $\overline{BD} \perp \overline{AC}$	o True o False	
10.	$\frac{AB}{AD} = \frac{BC}{BD}$	o True o False	